IMU

An IMU is a combination of sensors that measure orientation and motion with respect to an inertial reference frame.

Sensors in an IMU:

- 6 DOF IMU
  Contains 3 axis gyroscope and 3 axis accelerometer
- 9 DOF IMU
  Contains 3 axis gyroscope and 3 axis accelerometer and 3 axis magnetometer

Gyroscope: measure orientation angles and are angular rotation rate sensors. We integrate these rates to get angles but direct integration is not practical because of gyro bias and gyro drift. Even if the gyro is completely stationary on the desk it may display a rate that is constant with some irregularities and the constant is non zero this is known as gyro bias. When we integrate the rate wrt t and plot the graph it displays a linear graph with some irregularities and the angle is increasing this is known as gyro drift. Gyro bias causes Gyro drift. We can remove this using an extended kalman filter. The EKF will estimate the Gyro bias and then we correct the readings.

EKF: Linearizes the non-linear system model (kinematics) and measurement model (sensor readings) using Jacobian matrices (first-order Taylor series expansion).

Accelerometer: measure acceleration (linear, gravitational, centripetal, coriolis). Coriolis acceleration depends on the planet's rotation and our velocity (as it is w X v). We can even measure our tilt angle using the accelerometers and we call these angles roll and pitch angle. Like the gyros the integration of the acceleration drifts away due to some biases and we can use EKFs for that. We can supplement the GPS readings to help correct those velocity and position estimates back to normal through sensor fusion and using EKFs.

Magnetometer: a digital compass that measures the strength and direction of magnetic fields, particularly Earth's magnetic field, to provide an absolute heading reference. This helps to determine orientation relative to magnetic north and corrects for drift in the gyroscope, improving the overall accuracy of motion tracking when combined with accelerometers and gyroscopes.

Sensor fusion: By combining the magnetometer with accelerometers and gyroscopes an IMU can provide a comprehensive and more accurate understanding of an object's motion and orientation in 3D space.

Orientation:

- Roll Pitch and Yaw
  Roll: angle about longitudinal axis
  Pitch: angle about lateral axis
  Yaw: angle about vertical axis
  Gimbal Lock: when 2 axis align eg. when pitch rotates +-90 degrees the yaw and roll align

- Quaternions
  Quaternions are an extension of complex numbers that represent an axis-angle rotation in a compact, four-dimensional form. No gimbal lock. A unit quaternion q is represented by a scalar part and a 3D vector part:
  $q = w + xi + yj + zk$

IMU with ROS2:
1. Use a driver node to convert sensor data to ROS 2 messages
2. Identify the fields in these messages
3. Unit conversion if necessary
4. Publish the message to a topic
5. Define IMU's physical location and orientation relative to the rover's base
6. Perform Sensor Fusion